

Charged Black-Hole-Like Electronic Structure Driven by Geometric Potential of 2D Semiconductors

Bong Gyu Shin,* Ji-Hoon Park, Jing Kong, Soon Jung Jung,* and Young Jae Song*

One of the exotic expectations in the 2D curved spacetime is the geometric potential from the curvature of the 2D space, still possessing unsolved fundamental questions through Dirac quantization. The atomically thin 2D materials are promising for the realization of the geometric potential, but the geometric potential in 2D materials is not identified experimentally. Here, the curvature-induced ring-patterned bound states are observed in structurally deformed 2D semiconductors and formulated the modified geometric potential for the curvature effect, which demonstrates the ring-shape bound states with angular momentum. The formulated modified geometric potential is analogous to the effective potential of a rotating charged black hole. Density functional theory and tight-binding calculations are performed, which quantitatively agree well with the results of the modified geometric potential. The modified geometric potential is described by modified Gaussian and mean curvatures, corresponding to the curvature-induced changes in spin-orbit interaction and band gap, respectively. Even for complex structural deformation, the geometric potential solves the complexity, which aligns well with experimental results. The understanding of the modified geometric potential provides us with an intuitive clue for quantum transport and a key factor for new quantum applications such as valleytronics, spintronics, and straintronics in 2D semiconductors.

1. Introduction

The fundamental understanding of exotic objects and origins in the universe has been applied to solid state and condensed matter in various ranges, for example, the (massless) Dirac,^[1] Weyl,^[2] and Majorana^[3] fermions, the Big Bang theory,^[4] and black holes.^[5–8] The application of the fundamental particle theory has inspired many possibilities of the effective Hamiltonian of quasiparticles in the materials,^[9–12] which are tailored by their chemical composition, dimensionality, and symmetry along with quantum phenomena. In particular, a black hole, exhibiting the event horizon and singularity, is one of the most extreme objects observed in the universe,^[13,14] where the most advanced theories in the range of the macroscopic to microscopic scales encounter fundamental difficulties.^[15–17] Recent studies have probed the analogous black holes by acoustic and optical access, quantum Hall transport, and Bose-Einstein condensation.^[5–7,18]

Upon the realization of atomically thin 2D materials, the dimensionality and symmetry of the systems assign the emergence

of various electronic structures along with the exotic effective Hamiltonian for quasiparticles such as massless Dirac, Weyl, and Majorana fermions. In particular, the structural deformation in 2D materials is often responsible for governing emergent gauges to the effective Hamiltonian due to their flexibility, for example, the pseudomagnetic field.^[19–23] The structural deformation in 2D materials has been investigated in terms of curvature,^[24] but one fundamental issue regarding curvature remains, the geometric potential,^[25–28] which is an effective potential for constrained particles in curved 2D space concerning Riemannian geometry. The geometric potential still possesses unsolved fundamental questions such as the operator-ordering problem.^[27,28] The geometric potential has been demonstrated experimentally with an optical access,^[29,30] but remains to be observed directly at the atomic scale. The 2D materials are promising to be a stage for the geometric potential to comprehend the unsolved fundamental phenomena.

Here, we observed the curvature-induced ring-patterned bound states in structurally deformed monolayer MoS₂, which was confirmed to be driven by formulating the modified geometric potential. The modified geometric potential has a shallow

B. G. Shin, Y. J. Song
Department of Nano Science and Technology
Sungkyunkwan University (SKKU)
Suwon 16419, Republic of Korea
E-mail: shinbone@skku.edu; yjsong@skku.edu

B. G. Shin, Y. J. Song
SKKU Advanced Institute of Nanotechnology (SAINT)
Sungkyunkwan University (SKKU)
Suwon 16419, Republic of Korea

J.-H. Park, J. Kong
Department of Electrical Engineering and Computer Science
Massachusetts Institute of Technology
Cambridge, Massachusetts 02139, USA

S. J. Jung
Max Planck Institute for Solid State Research
Heisenbergstrasse 1, 70569 Stuttgart, Germany
E-mail: s.jung@fkf.mpg.de

Y. J. Song
Department of Nano Engineering
Sungkyunkwan University (SKKU)
Suwon 16419, Republic of Korea

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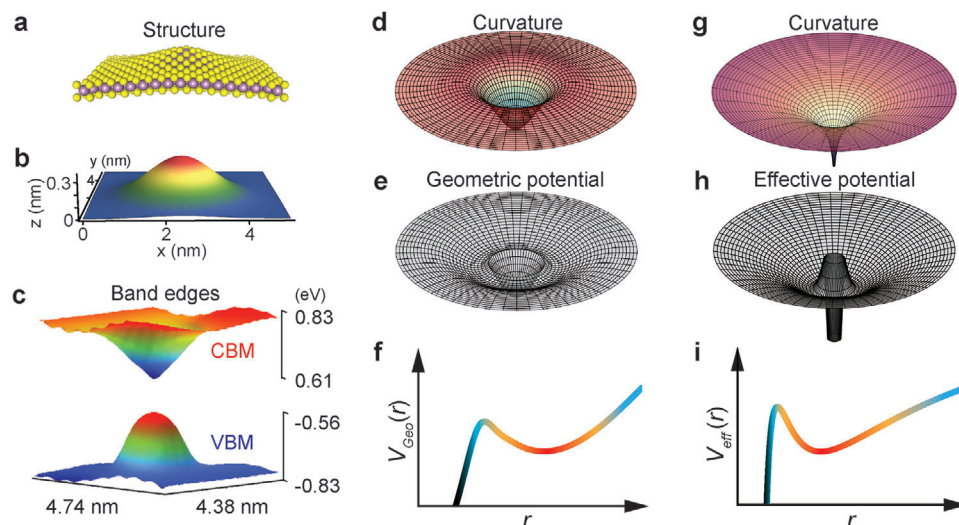


Figure 1. Geometric potential in 2D semiconductors. a) Structural model of Gaussian-like deformed monolayer MoS₂. b) Surface from Gaussian fitting of (a). c) the local conduction band minimum (CBM) and valence band maximum (VBM) extracted from the density functional theory result. d–f) Curvature (d), geometric potential (e), and radial line profile of the geometric potential (f) in the Gaussian-like deformed monolayer MoS₂. g–i) The curvature of space (g), the effective potential (h), and radial line profile of the effective potential (i) near a black hole. The shallow depth in the geometric potential of the deformed MoS₂ allows bound states near the center of the deformation, analogous to the effective potential of the black hole. The shallow depths in the geometric and effective potentials ((f) and (i)) are remarked by red and orange colors.

depth near the high curvature region, which is an analog of the effective potential of a rotating charged black hole. The ring-shaped bound states were located at the shallow depth region of the modified geometric potential, analogous to the stable orbits outside the black hole. The bound states exhibited a larger angular momentum with a larger radius of the ring-patterned bound states outside the center of the high curvature region. The density functional theory (DFT) and tight-binding calculations with deformed structures were well-matched with the results of the modified geometric potential, quantitatively. We confirmed the validity of the geometric potential even in the complex structural deformation, which agrees well with the experimental results. The modified geometric potential is related to the curvature-induced changes in both spin-orbit interaction and band gap. The modified geometric potential provides an intuitive picture of the curvature effect in the electronic structure of 2D semiconductors for ideal applications of valleytronics, spintronics, and straintronics.

2. Results and Discussion

Figure 1a shows the structural model of Gaussian-like deformed monolayer MoS₂ and the Gaussian fitting result of Figure 1a is displayed in Figure 1b. The electronic structure of the structural model was calculated based on DFT. In Figure 1c, the band edges (the local conduction band minimum (CBM) and valence band maximum (VBM)) were extracted from the DFT results. In the deformation of MoS₂, the local band gap decreases at high curvature regions, which is confirmed by our previous studies.^[24] The origin of the surface deformations in monolayer MoS₂ on SiO₂ substrate is the surface roughness of the substrate. The surface deformation induces bending strain up to ≈6%. The previous studies show that the curvature-induced band gap fluctuation exhibits a linear relation between band gap change (or band edge

change) (ΔE_C) and the mean value of the absolute principal curvatures (H^*), $\Delta E_C = 2\alpha H^*$, where α is the proportional coefficient ($\alpha = -4.238$ eV nm, from the previous study^[24]). The relation means that the local band gap change is proportional to the bending strain. It is worth noting that the curvature-induced band gap variation induces charge localization, which causes metal-insulator transition with the critical doping concentration when the structural disorder is involved.^[31] Our concern is the effective potential (Hamiltonian) model for charge carriers through the curvature formation in 2D semiconductors, semiclassically. One of the direct considerations is to investigate the effective potential. From the viewpoint of charge carriers inside the 2D semiconductors, the local band edge fluctuation ($\Delta E_C/2$) and the confinement by the material itself as a curved space determine where the energy states are possible to live. The confining effect of curved 2D systems is simplified by the geometric potential, $V_g = -(\hbar^2/8m)(k_1 - k_2)^2 = -(\hbar^2/2m)(H^2 - K)$, where $\hbar$, m , and (k_1, k_2) are the reduced Planck constant, effective mass, and set of principal curvatures, respectively, which is derived by the thin-wall quantization.^[26] The combination of the two effects in terms of potentials can be written by $V_{Geo} = V_g + \Delta E_C/2$. The expression of V_{Geo} can be in the same form as the geometric potential (V_g) of the confined free carriers in the curved 2D system when we introduce the modified mean and Gaussian curvatures. The definitions of the modified mean and Gaussian curvatures depend on the form of ΔE_C that is extracted from the experimental results. Based on the previous experimental results,^[24] we introduce the modified geometrical potential in 2D semiconductors with modified mean and Gaussian curvatures ($\bar{H}$ and $\bar{K}$, respectively) as $V_{Geo} = 4\beta(\bar{H}^2 - \bar{K})$, where β is $-(\hbar^2/8m)$, $\bar{H} = \frac{|k_1| + |k_2|}{2} + \frac{\alpha}{8\beta}$, and $\bar{K} = \frac{|k_1||k_2|(1 + \text{sign}(k_1 k_2))}{2} + (\frac{\alpha}{8\beta})^2$, where $\text{sign}(x) = 1$ if $x \geq 0$, -1 if $x < 0$ (See the detail in Supporting Information). It

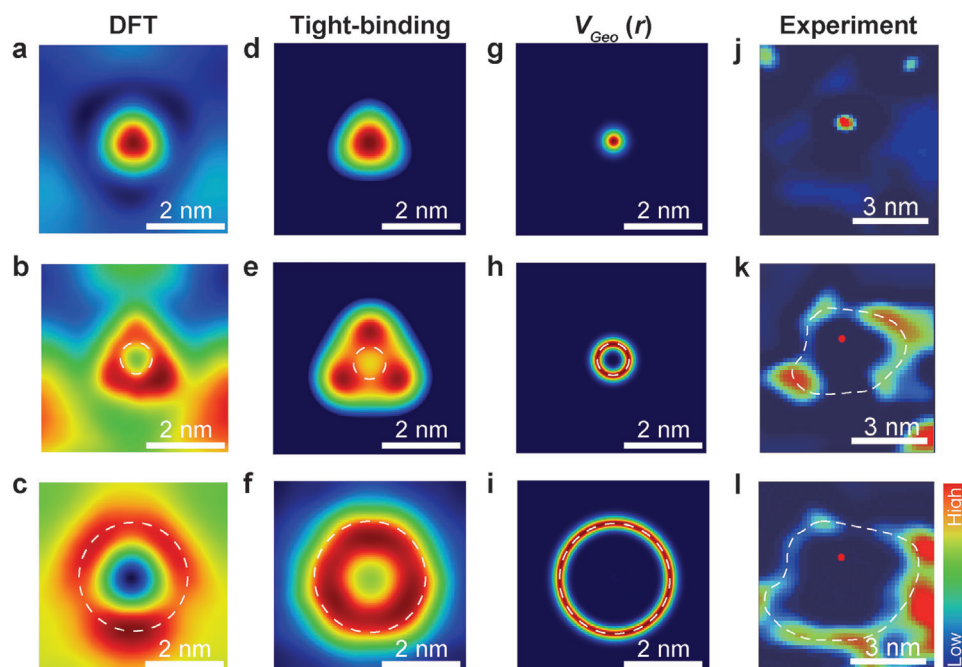


Figure 2. Curvature-induced geometric potential and bound states in monolayer MoS₂. a–c) DFT calculation results of bound states for holes near VBM (at 0 (a), –0.11 (b), and –0.4 (c) eV) with the structural model of spherical curvature for monolayer MoS₂ (Figure 1a). d–f) Tight-binding calculations result with the fitted Gaussian surface of Figure 1b (at 0 (d), –0.09 (e), and –0.45 (f) eV). g–i) The modified geometric potential results from the Gaussian surface (Figure 1b) as in the tight-binding results (at 0 (g), –0.10 (h), and –0.41 (i) eV). The tight-binding and geometric potential results are in good agreement with the DFT results. j–l) STS mapping results near bump morphology in monolayer MoS₂ at biases of –1.98 (near VBM) (j), –2.1 (k), and –2.13 (l) eV, respectively. The red dot indicates the center of the bump. The energy of each is relative to the 0 eV (VBM), which defines the ground-bound state for holes (a,d,g). The electron cases near CBM are similar to the hole cases except for the sign of bound state energy (lower energy means stronger binding due to that the electronic structures were defined by electron states). Color-scale bars are displayed at the side of (l) for all.

is noteworthy that if charge doping is included, the curvature-bandgap relation should concern each band edge of the conduction and valence, separately, due to the doping-induced band edge flattening as confirmed in the previous study.^[31] Therefore, in general, the α is also a function of doping concentration for the conduction and valence band edges. Figure 1d shows the curvature H^* from the Gaussian shape of MoS₂ (height of 1 nm and half width at half maximum (HWHM) of 4.7 nm) and the corresponding geometric potential is displayed in Figure 1e,f. The geometric potential (Figure 1e) shows shallow depth, which provides stable orbit states near the center of the deformation. The formation of geometric potential is analogous to the effective potential of a black hole. The curvature of the spacetime driven by a black hole (Figure 1g) leads to the effective potential with a shallow depth of the potential (Figure 1h,i), which exhibits a stable orbit near the black hole.

From the formulation of the geometric potential, the stable bound states near the deformation center are expected as rim patterns. To investigate the bound states of the Gaussian deformation case, we performed DFT and tight-binding calculations and solved eigenstates of the Schrödinger equation with the geometric potential for the same Gaussian morphology (height of 0.36 nm and HWHM of 1.47 nm) (Figure 2). The DFT results in the valence band side (for holes) show bound states of relative energies to VBM (defined as 0 eV); 0 eV, –0.11, and –0.4 eV in Figure 2a–c, respectively. At the center of the Gaussian morphology, a point-like bound state is located as a ground-bound

state (Figure 2a). In the high curvature region near the center, the bound state of higher binding energy (–0.11 eV) is formed into a ring shape (Figure 2b). An outer stable orbit of the higher binding energy (–0.4 eV) is found away from the center (Figure 2c). The tight-binding results (Figure 2d–f) align well with DFT results, quantitatively, in terms of shape and energy values of each bound state (0 (d), –0.09 (e), and –0.45 (f) eV). The bound states with the modified geometric potential (V_{Geo}) are also in good agreement with the DFT and tight-binding results (Figure 2g–i). The only small difference is that the bound states from the modified geometric potential show ideal circular shapes, while the DFT and tight-binding results involve a three-fold symmetry of protrusions. The deviation arises from the continuum approximation limit due to the discreteness of the lattice in both DFT and tight-binding results. The modified geometric potential is under the continuum approximation in the semiclassical approach. Such a consistent result among the different theoretical approaches guarantees the validity of the modified geometric potential as an effective Hamiltonian for curvature effects in deformed 2D semiconductors. Experimental observation by scanning tunneling spectroscopy (STS) demonstrated circular patterns of bound states on the bump morphology of MoS₂ (Figure 2j–l). At the higher binding energy, the circular pattern became larger, which is consistent with theoretical results. In particular, the dot-like bound (localized) state at the deformation center (high curvature region) is not allowed by the geometric potential (V_g),^[32] while the modified geometric potential (V_{Geo}) captures the dot-like bound

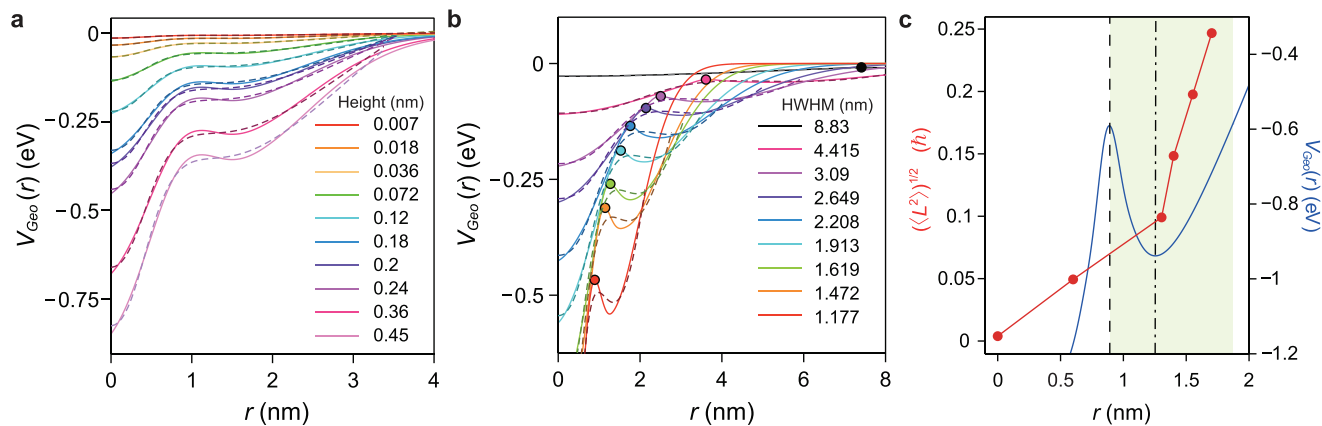


Figure 3. Variations in the geometric potential and the angular momentum of the ring-patterned bound states. a) Geometric potentials from height change in Gaussian deformed monolayer MoS₂ at a fixed radius of the deformation (HWHM = 1.472 nm). b) Geometric potential changes as a function of HWHM in the Gaussian deformation with the fixed height of 0.36 nm. Dashed lines for each case are fitted by the black-hole-like potential. c) Characterization of the angular momentum ($(\langle L^2 \rangle)^{1/2}$) for ring patterns of bound states from the modified geometric potential (height = 0.36 nm, HWHM = 1.225 nm). For the angular momentum (red dots, the left y-axis), r on the x-axis means the radius of the bound state ring patterns. The right y-axis corresponds to the modified geometric potential (blue curve). The dashed and dash-dot vertical lines indicate the local maximum and the local minimum in the modified geometric potential. The green shaded area shows the bound region for stable orbits.

state at the center of the deformation, which matches the experimental results. In addition, the ring-like bound states in the geometric potential exhibited angular momentum, implying that the ring-like bound states through the modified geometric potential have also angular momentum (as described later). Furthermore, curvature-induced change in the spin-orbit interaction governs the pseudo-magnetic field proportional to the Gaussian curvature.^[19] Across a curvature region, propagating wavefunctions will gain changes associated with the Berry-spin connection as a scattering process,^[19] implying a new way to applications of curvature-induced wavefunction interference. As a result, the modified geometric potential includes the two parts of the curvature effect for changes in both spin-orbit interaction (the modified Gaussian curvature) and band gap (the modified mean curvature) in 2D semiconductors such as transition metal dichalcogenides, which warrants future investigation for the quantum dynamical behavior of wavefunctions through the modified geometric potential of desired deformed structures. Note that quantum states of electrons by the curvature-induced modified geometric potential in the conduction band side behave in the same mechanism on the valence band side of holes (Figure S1, Supporting Information).

Figure 3 shows the variation of the modified geometric potential as a function of changes in height and radius ($\approx$ HWHM) of the Gaussian morphology of MoS₂. When the HWHM of the Gaussian morphology is fixed, the change in height of the morphology affects the flatness of the modified potential (Figure 3a). The smaller change in height means a flatter morphology with lower curvature, resulting in a flatter potential. The local maximum point does not vary significantly with height change, which is close to the HWHM, characterizing an effective radius of the high curvature region in the Gaussian morphology. On the other hand, when the height of the Gaussian morphology is fixed, larger HWHM leads to the flattening of the modified potential due to the formation of lower curvature (Figure 3b). The local

maximum point is located at almost $\text{HWHM}/\sqrt{2\ln 2}$, indicated by colored dots for each case in Figure 3b. Each result in Figure 3 was fitted by the generalized effective potential of black holes in

the form of $V_b(r, d) = \sum_{i=0}^4 v_i / (r^2 + d^2)^{i/2}$, where v_i is a coefficient and d control the singularity of the potential as a vertical position of the black hole to the surface of the 2D system. The fitting function originates from the case of Kerr-Newman spacetime for a rotating charged black hole.^[33] The fitting coefficients correspond to the physical parameters of a black hole, such as mass, angular momentum, and charge, characterizing the unique properties of the black hole as expected from the uniqueness theorem of black holes^[34] (See more details in Supporting Information). Note that the surface deformation itself does not induce charge, and therefore the “embedded” charge as a black hole source should be outside of the “2D universe” in the extra-dimension at the distance of d normal to the 2D universe. This “dark charge” is an analog of the dark matter but with Coulomb interactions. (However, we are open to the true 2D potential form if the embedded charge is very close to the 2D universe. There are several approaches to constructing a 2D potential.^[35] A 2D potential form has a logarithmic function with singularity issues.) From Gauss’ law, we can estimate the dark charge (or mass) density ρ , $\nabla^2 V_b = \rho$ in an appropriate unit. When applying Gauss’ law only to the volume of the 2D universe, the dark charge density becomes zero due to the neutrality of the structural deformation (without charge doping), according to the charge conservation laws in the 2D system. Since the structural deformation is induced by external factors such as the substrate or any other contact materials, the dark charge is not free even for its virtuality. In addition, structural deformation is static, which should have a stationary black hole solution. The Majumdar-Papapetrou solution is a well-known example of stationary solutions for multiple charged black holes, where each black hole has a charge (q_i) equal to its mass (m_i), $|q_i| = \kappa m_i$ for

the i -th black hole, where κ is a constant in an appropriate unit, in the equilibrium between gravitational attraction and electrostatic repulsion^[27–29,36,37] (See details in Supporting Information). Therefore, the dark charge counts as dark mass (matter), directly. If charge doping is applied, the previous results show that localized charges over the high curvature regions induce local band bending toward the band edge flattening.^[31] The band edge flattening corresponds to the weakening of the modified geometric potential, that is, the screening of the dark charge by the localized charge. This doping effect is analogous to the evaporation of a black hole by Hawking radiation.^[15] We conjecture that doping concentration corresponds to the “time of the 2D universe” and dark charge (or dark mass) is evaporating as a function of doping concentration, the analog of time. The correspondence between doping concentration and time of the 2D universe is also triggering the scale factor for metrics as analogous to the Hubble expansion in the universe. From Einstein’s field equations with the positive cosmological constant (Λ), the scale factor $a(t)$ diverges to zero as time t becomes infinite, leading to the flat universe.^[38] For our 2D universe, the scale factor for Hubble expansion is a function of doping concentration. When applying a higher doping level, eventually the 2D universe becomes flat, which coincides with the localization-delocalization transition or metal–insulator transition with the critical doping concentration in 2D semiconductors.^[31,39] Briefly, the modified geometric potential as a charged black hole analog involves two mechanisms for the evolution; mass evaporation and Hubble expansion, both as a function of doping concentration. This conjectures that doping concentration is an analog of time in the “2D universe” and the cosmological constant is determined by the strength of band edge flattening via doping. (For further discussion, see Supporting Information). The exact formulation for each physical factor warrants further study.

Although the analytical formulation is not fully achieved for mass evaporation and scale factor, numerical results for the cosmological constant and power factor of mass evaporation evolution can be estimated from the band edge flattening via doping calculated by DFT. The renormalized band edge fluctuation range (ΔE) is defined by the difference between the maximum (E_{max}) and minimum (E_{min}) in the band edge fluctuations over the spatial space, $\Delta E(\Delta n_e) = (E_{max}(\Delta n_e) - E_{min}(\Delta n_e))/E_0$, where E_0 is the band edge fluctuation range at zero doping and Δn_e is electron doping concentration. This rescales the global energy scale of the band edge fluctuations via doping, which directly rescales the modified geometric potential itself via the curvature-band gap (edge) relation. From Gauss’ law, mass evaporation by Hawking radiation, and empty Lemaître model (assumed low mass density and flat curvature) along with the co-moving frame (Robertson-Walker metric), we can estimate that $M_{bending,0}/a(\Delta n_e) \approx \Delta E(\Delta n_e)$, obtained from $M_{bending,0} \approx (\Delta n_e)^{1/3}$ and $a(\Delta n_e) \approx \exp(\sqrt{(\Lambda/3) \cdot c \cdot b \cdot \Delta n_e})$, where $M_{bending,0}$ is the dark mass at the neutral state without doping, b is a constant for a unit, c is the analog of the speed of light (Figure S4, Supporting Information). By fitting the band edge fluctuation range $\Delta E(\Delta n_e)$ with $(\Delta n_e)^{1/3} \exp(\sqrt{(\Lambda/3) \cdot c \cdot b \cdot \Delta n_e})$, we evaluated the cosmological constant analog in the 2D universe, $0.0187/(bc)^2$ for the valence band edge and $0.0044/(bc)^2$ for the conduction band edge. The constant c is conjectured to be related to the temperature by $c \approx \sqrt{T_c}$,

where T_c is the critical temperature for thermally activated percolation as we discussed in Supporting Information. Although we still need to determine b for a unit, the positive value of the cosmological value and the well-fitting 1/3 power for mass evaporation indicate that the doping-induced flattening of the metric (analog of the Hubble expansion in the 2D universe) and the charged black hole analog in three dimensions, respectively, are as we hypothesized. Therefore, localization-delocalization transition or 2D metal–insulator transition in structural disorder aligns well with the analogs of the physical concepts in black holes. It is worth noting that the analog of black hole physics in deformed 2D semiconductors is not the miniaturization of a black hole as like a microcosm reflecting or representing the macrocosm of the universe.

Figure 3c shows the angular momentum of ring-patterned bound states as stable orbits near the center of the Gaussian structural deformation. At the center of the Gaussian structural deformation, dot-patterned and ring-patterned bound states in the local density of states maps (Figure S2, Supporting Information) exhibited lower angular momentum with a smaller radius. Outside the bump (local maximum) of the modified geometric potential, the shallow depth of the modified geometric potential (green shaded region in Figure 3c) allowed stable orbits with larger angular momentum. The stable orbits started at the local minimum of the potential. The larger radius of the ring-shaped bound state at the rim of the Gaussian deformation shows larger angular momentum, which is in good agreement with previous theoretical results of the geometric potential in Gaussian structural deformation of a 2D system.^[32] Notably, the energy-angular momentum relationship exhibits linearity with equally spaced quantized energy levels, which is not typical of classical systems (Figure S3, Supporting Information).

Figure 4 demonstrates the robust capability of the modified geometric potential in a complex morphology. The surface morphology in monolayer MoS₂ on SiO₂ was measured by scanning tunneling microscopy (STM) (Figure 4a). The STS mapping result in the same region as Figure 4a shows the complex rim patterns near the protrusions of the deformed MoS₂ (Figure 4b). In Figure 4c, the tight-binding result with the same morphology as Figure 4a reproduced the complex rim patterns, which aligned well with the experimental result. With the same morphology, the Schrödinger equation with the modified geometric potential again demonstrated the complex rim patterns (Figure 4d), confirming the robust capability of the modified geometric potential to solve the electronic structures in a complex deformed structure of 2D semiconductors. Furthermore, in a relatively simple Gaussian-like deformation with a skewness of the shape, it is also expected to manifest various ring-like patterns toward future work. Such a validity of the modified geometric potential to the electronic structures in structural disorder is directly linked to the exciton formations. Indeed, the charged exciton formations assisted by the curvature effect were found in photoluminescence mapping results.^[40] Other physical factors, such as electrical confinement, magnetic field, effective mass, etc., are also accompanying with the curvature effects for exciton formation, which opens a promising route to unveil the underlying mechanism.^[41–43]

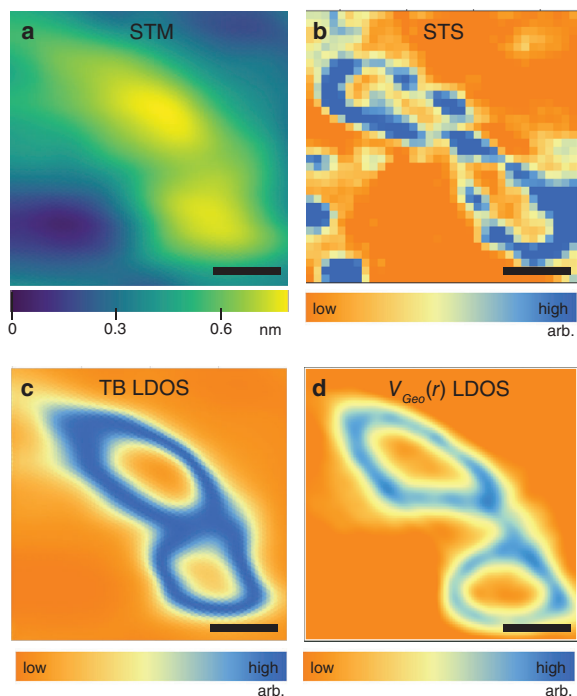


Figure 4. Complex rim patterns of bound states. a) Topological STM image. b) STS mapping of the same area in (a) at the bias of 0.43 V below VBM. c,d) Tight-binding (TB) (c) and the modified geometric potential ($V_{Geo}(r)$) (d) calculation results, corresponding to the structure of (a). All Scale bars of each plot indicate 5 nm. Both TB and the modified geometric potential results align well with the experimental results.

3. Conclusion

In summary, we observed the bound states of dot and ring patterns in/near the high curvature regions using STM/S, which was demonstrated by the modified geometric potential as the curvature effect in the electronic structure of 2D semiconductors. The modified geometric potential manifests a shallow depth near the high curvature region, which resembles the effective potential of a rotating charged black hole exhibiting stable orbits outside the black hole. The stable orbits in the shallow depth of the modified geometric potential exhibit a larger angular momentum with a larger radius of the ring-shaped bound states outside the center of the high curvature region. The theoretical calculations with DFT, tight-binding, and the modified geometric potential were in good agreement with each other, confirming the validity of the modified geometric potential for solving a complex structural deformation in 2D semiconductors, which aligns well with the experimental results. The modified geometric potential is related to the curvature-induced changes in both band gap and spin-orbit interaction, which provides a key concept for valley and spin transport toward the spintronics, valleytronics, and straintronics in 2D semiconductors.

4. Experimental Section

Sample Preparation: Monolayer MoS_2 was grown by low-pressure metal-organic chemical vapor deposition.^[44] Molybdenum hexacarbonyl

($\text{Mo}(\text{CO})_6$, Sigma Aldrich) and diethyl sulfide ($(\text{C}_2\text{H}_5)_2\text{S}$, Sigma Aldrich) were prepared as precursors of Mo and S, respectively. They were introduced into a 1-in quartz tube furnace in the gas phase together with Ar carrier gas. The MoS_2 film was synthesized on a SiO_2 (300 nm)/Si wafer with a flow rate of 100 SCCM (Standard Cubic Centimeter per Minute) for Ar, 0.6 SCCM for $\text{Mo}(\text{CO})_6$, and 2.0 SCCM for $(\text{C}_2\text{H}_5)_2\text{S}$ at a growth temperature below 350 °C for 15 h. After growth, the sample was cooled down to room temperature. For the transfer process, Poly(methyl methacrylate) (PMMA) was spun onto as-grown MoS_2 on SiO_2 /Si at 1000 rpm. The PMMA-coated sample was dipped into a 2 M KOH solution to detach the PMMA/ MoS_2 film from the SiO_2 substrate. The PMMA/ MoS_2 film was cleaned out by floating on the distilled water three times and, subsequently, the film was transferred onto a target substrate. The PMMA layer was removed by soaking the sample in acetone and isopropyl alcohol baths in order.

Scanning Tunneling Microscopy and Spectroscopy: STM and scanning tunneling spectroscopy (STS) were performed on transferred monolayer MoS_2 on SiO_2 /Si wafers with Au or Pd electrode contact to the MoS_2 at ≈ 4.8 K under ultra-high vacuum (UHV) of $\approx 10^{-10}$ mbar using home-built STM. A chemically-etched W tip was used as an STM probe after removing oxidation layers by e-beam heating. For STS, the conventional lock-in techniques were conducted with an oscillation frequency of 417 Hz and an amplitude of 7 mV. All samples were thermally cleaned out below 380 °C in the UHV condition.

Tight-Binding Methods: The Gaussian-shape structural deformation of monolayer MoS_2 was modeled by the height change of lattice points. The tight-binding method within the third-nearest-neighbor was performed based on the three-band model with d -orbital (d_{z^2} , d_{xy} , and $d_{x^2-y^2}$).^[45] In the deformed structures, the strain-induced changes of hopping parameters t_{ij} between i and j sites were considered as $t_{ij} = t_{ij}^0 \exp(-\beta_t(a_{ij}/a_0 - 1))$, where a_0 , t_{ij}^0 , and a_{ij} were unstrained lattice distance, primitive hopping parameter, and strained distance between i and j sites, respectively. The β_t factor was empirically applied by 5 (Ref. [46]). Due to the simple approximation of the structural deformation considering only height fluctuations, the authors applied the additional terms of height fluctuation coupling between two positions up to the third nearest neighbors to mimic the localization near the band edges toward the high curvature center observed in the experimental results (See the details in Supporting Information).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

2D semiconductors, band gap, black holes, curvatures, geometric potentials

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